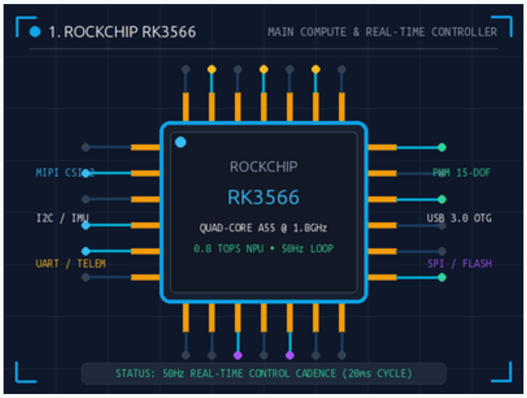
■ MICRODUCK PHYSICAL AI MASTERCLASS

PHASE 1 HANDOUT

Module 1 Hardware Architecture: Sensor-Actuator Stack, 50Hz Real-Time Loop, & Silicon Constraints

MuJoCo + ONNX

COMPUTE: Rockchip RK3566	**CADENCE: 50 Hz (20ms / step)**	**DEGREES OF FREEDOM: 15 Actuators**	**AI STACK: PyTorch PPO → ONNX**



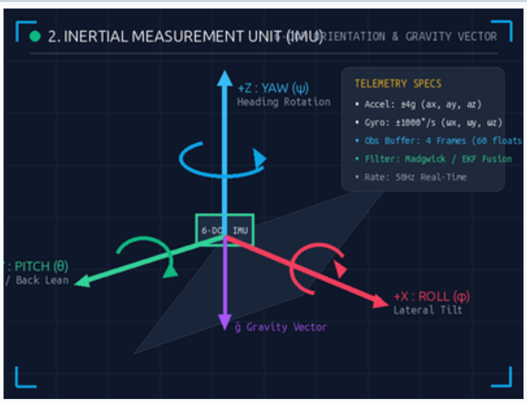
1. The Brain: Rockchip RK3566

ONBOARD COMPUTE & REAL-TIME KERNEL

The Core Engine: A quad-core ARM Cortex-A55 @ 1.8GHz with an integrated 0.8 TOPS NPU running an optimized Linux OS. It serves as the physical robot's central nervous system.

The 12-Year-Old Analogy: Like an orchestra conductor who doesn't play the instruments but cues 15 musicians exactly 50 times every single second.

- **50Hz Control Cadence**: Strict 20ms execution loop managed by real-time daemon (robotd).
- **Interface Peripherals**: MIPI-CSI2 camera, high-speed UART IMU, CAN/PWM motor buses.
- **Edge Inference**: Executes pruned ONNX Actor graph with <2.8ms latency per step.



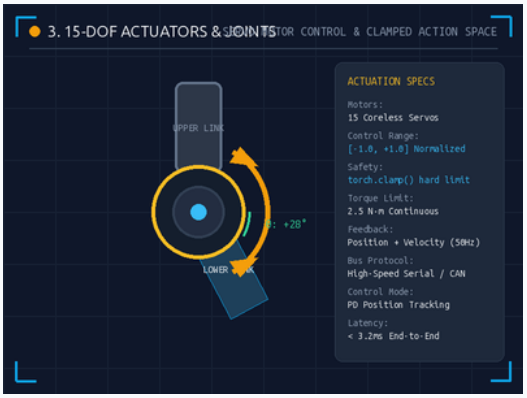
2. The Inner Ear: 6-DOF IMU Sensor

INERTIAL NAVIGATION & GRAVITY VECTOR

The Vestibular Sense: Combines a 3-axis accelerometer and 3-axis gyroscope to track Pitch (θ), Roll (ϕ), and Yaw (ψ) along with dynamic gravity vectors.

The 12-Year-Old Analogy: Close your eyes and stand on one leg. Your inner ear tells you instantly if you are tilting before your eyes even notice.

- **Observation Window**: Fixed sliding buffer of 4 frames × 15 sensors = 60 float telemetry tensor.
- **Sensor Fusion**: Real-time Madgwick / Extended Kalman Filter running at 200Hz internal rate.
- **RL Input**: Normalizes angular velocity [ωx, ωy, ωz] and projected gravity vector [gx, gy, gz].



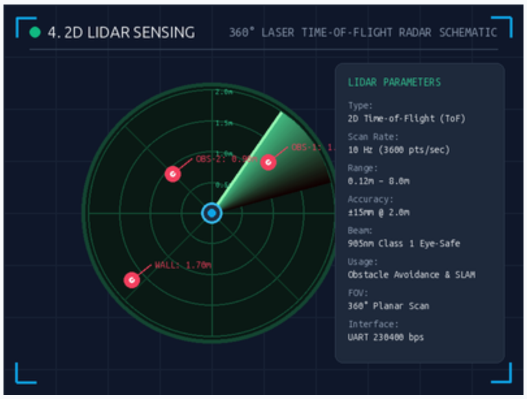
3. The Muscles: 15-DOF Actuators

ROBOTIC JOINTS & ACTION CLAMPING

High-Torque Articulation: 15 coreless servo motors drive hips, knees, and ankles. The neural network outputs normalized action targets.

The 12-Year-Old Analogy: Muscles have bone stops so your knee can't bend backwards. We clamp the AI math in silicon so it never breaks a joint.

- **Action Space**: Strictly bounded a ∈ [-1.0, 1.0] via torch.clamp() in ONNX graph.
- **Control Protocol**: High-speed serial bus delivering target velocity and PD torque gains.
- **Joint Limits**: Enforced in both URDF/MJCF definitions and firmware safety watchdogs.



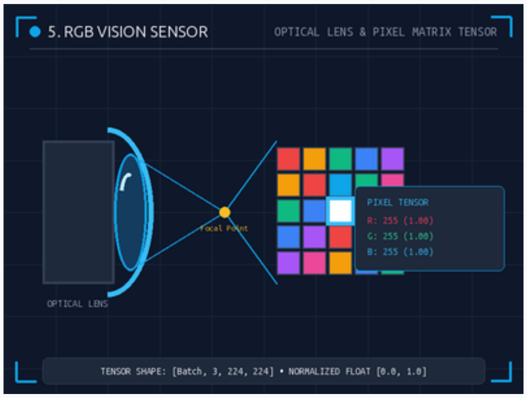
4. The Bat Sonar: 2D LiDAR Scanner

PLANAR RANGEFINDING & SPATIAL SLAM

Time-of-Flight Ranging: Emits 905nm pulsed laser rays at 10Hz, capturing 360° point clouds for real-time obstacle avoidance and hallway navigation.

The 12-Year-Old Analogy: Exactly like a bat navigating a pitch-black cave using ultrasonic clicks, but using light beams that bounce off walls at the speed of light.

- **Effective Range**: 0.12m to 8.0m with ±15mm distance precision at walking height.
- **Spatial Perception**: Generates polar obstacle distance map to modulate locomotion goals.
- **Sensor Interface**: Direct DMA UART link feeding 3600 spatial measurement points per second.



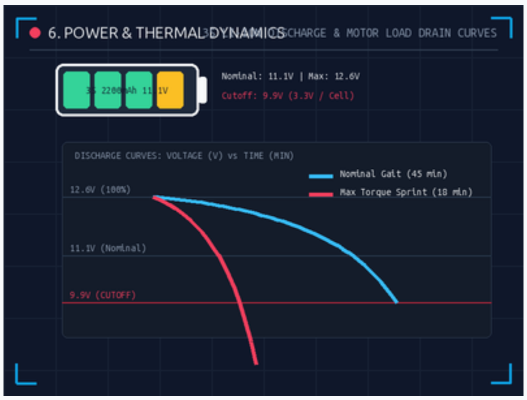
5. The Visual Cortex: RGB Camera

OPTICAL SENSOR & TENSOR PROCESSING

Embodied Vision: Captures raw optical photons through an ultra-wide lens and translates them into an RGB pixel matrix tensor for neural feature extraction.

The 12-Year-Old Analogy: A computer doesn't see a picture; it sees a giant grid of Red, Green, and Blue numbers between 0 and 255.

- **Tensor Pipeline**: Preprocessed into [Batch, 3, 224, 224] float tensors normalized to [0.0, 1.0].
- **Hardware Acceleration**: Direct hardware ISP color conversion passing directly into RK3566 NPU.
- **Semantic Guidance**: Supplements high-frequency blind proprioception with terrain recognition.



6. The Gas Tank: Power & Thermal Dynamics

3S LI-ION DISCHARGE & LOAD BUDGETING

Energy Limitations: A 3S 11.1V 2200mAh Lithium-Ion battery powers all 15 actuators. Motor current draw directly determines operating duration.

The 12-Year-Old Analogy: If you sprint as fast as you can, you will collapse in 1 minute. If you jog steadily, you can run for an hour. The AI must budget energy.

- **Voltage Thresholds**: 12.6V Full Charge → 11.1V Nominal → 9.9V Emergency Low-Voltage Cutoff.
- **Gait Efficiency**: Smooth 50Hz policies consume ~3.2A (45 min runtime) vs 15A during erratic thrashing.
- **Reward Regularization**: RL training introduces energy penalty term: reward_energy = -c * ||torque||².

■ PHYSICAL AI PHASE 1 ARCHITECTURE SUMMARY & NEXT STEPS

- **MuJoCo Simulation:** Train policy in headless virtual sandbox where falling costs $0.
- **Hardware Clamping:** Bake torch.clamp() directly into ONNX graph before exporting to edge.
- **Phase 2 Transition:** Proceed to Module 2 Reinforcement Learning with PPO and Gym environments.

■ KNOWLEDGE CHECK

ASSESSMENT

Test your understanding of the Microduck hardware before moving to the software layer.

Phase 1 Review

Q1: Control Frequency & Cadence

MULTIPLE CHOICE

Why does the Rockchip brain need to send motor commands 50 times a second?

- **A.** To keep the battery charged
- **B.** To constantly catch its balance so the duck doesn't fall
- **C.** Because Linux only runs at 50Hz

Q2: Actuator Clamping & Protection

MULTIPLE CHOICE

What happens if the AI sends a motor command of 2.5 (past the 1.0 limit)?

- **A.** The robot walks faster
- **B.** The IMU takes over
- **C.** The physical motor breaks

Q3: Sensory Perception in Darkness

MULTIPLE CHOICE

If the duck is in a completely dark room, how does it see a wall?

- **A.** RGB Camera night vision
- **B.** LiDAR shoots invisible lasers
- **C.** The inner ear feels it

Answer Key: 1-B, 2-C, 3-B